

JEE MAIN REVISION MASTER: ALTERNATING CURRENT

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Series LCR Circuit & Impedance	11 PYQs	Circuits
High	AC Fundamentals: Peak, RMS & Phase	9 PYQs	Foundations
Moderate	Resonance, Q-Factor & Bandwidth	6 PYQs	Resonance
Moderate	Reactance: Inductive & Capacitive	5 PYQs	Impedance Elements
Moderate	LC Oscillations & Transformers	4 PYQs	Devices & Oscillations
Standard	Power in AC Circuits & Power Factor	1 PYQs	Power & Efficiency

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Series LCR Circuit & Impedance	$Z = \sqrt{R^2 + (X_L - X_C)^2}$, $\tan\phi = \frac{X_L - X_C}{R}$, $V_{\text{total}} = \sqrt{V_R^2 + (V_L - V_C)^2}$
AC Fundamentals: Peak, RMS & Phase	$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$, $I_{\text{avg, half}} = \frac{I_0}{2\pi}$
Resonance, Q-Factor & Bandwidth	$\omega_0 = \frac{1}{\sqrt{LC}}$, $Z_{\min} = R$, $I_{\max} = \frac{V}{R}$, $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}}$
Reactance: Inductive & Capacitive	$X_L = \omega L = 2\pi f L$, $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
LC Oscillations & Transformers	$V_s V_p = N_s N_p = I_p I_s$, $\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p}$, $q(t) = q_0 \cos(\omega_0 t)$
Power in AC Circuits & Power Factor	$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$, $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$, $I_{\text{wattless}} = I_{\text{rms}} \sin\phi$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Series LCR Circuit & Impedance (High Yield — 11 PYQs)

Core Principle: Series LCR impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$. Circuit current $I = (V / Z)$. Phase angle $\tan\phi = (X_L - X_C / R)$. Net voltage phasor $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Caution / JEE Pitfall:** Voltages across series AC elements do NOT add algebraically: $V_{\text{(total)}} \neq V_R + V_L + V_C$! Always use phasor vector addition: $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Speed Trick:** If $X_L > X_C$, circuit is inductive (voltage leads current by ϕ). If $X_C > X_L$, circuit is capacitive (current leads voltage by ϕ).

Representative Source PYQs:

- **[2025 Shift] Q#2 (MCQ):** "A series LCR circuit containing an AC source of 100 V has an inductor and a capacitor of reactances 24Ω and 16Ω respectively. ..." **[Ans: N/A]**
- **[2024 Shift] Q#1 (MCQ):** "The figure shows variation of R, X_L and X_C with frequency f in a series L-C-R circuit. Then, for what frequency point is the ci..." **[Ans: N/A]**
- **[2024 Shift] Q#3 (MCQ):** "Which of the following combinations should be selected for better tuning of an L-C-R circuit used for communication?..." **[Ans: N/A]**

3.2 AC Fundamentals: Peak, RMS & Phase (High Yield — 9 PYQs)

Core Principle: Sinusoidal AC quantities: $I_{\text{(rms)}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$, $V_{\text{(rms)}} = \frac{V_0}{\sqrt{2}}$. Average current over full cycle is 0; average over half cycle is $I_{\text{(avg, half)}} = (2I_0 / \pi) \approx 0.637 I_0$.

■ **Caution / JEE Pitfall:** AC voltmeters and ammeters measure RMS values, not peak values! When JEE specifies '220 V AC', it always denotes $V_{\text{(rms)}}$ (peak voltage is $V_0 = 220\sqrt{2} \approx 311$ V).

■ **Speed Trick:** Instantaneous power $p(t) = v(t)i(t)$ fluctuates at double frequency (2ω) about average power $P_{\text{(avg)}} = V_{\text{(rms)}}I_{\text{(rms)}}\cos\phi$.

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "A light bulb rated 100 W is connected to an AC source of 220 V, 50 Hz. The rms current through the bulb is..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "In a circuit containing a pure resistor connected to an AC source,..." **[Ans: N/A]**
- **[2025 Shift] Q#1 (MCQ):** "A sinusoidal voltage produced by an AC generator at any instant t is given by an equation $V = 311\sin 314 t$. The rms value of voltage and freq..." **[Ans: N/A]**

3.3 Resonance, Q-Factor & Bandwidth (Moderate Yield — 6 PYQs)

Core Principle: At resonance: $X_L = X_C$, resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ ($f_0 = \frac{1}{2\pi\sqrt{LC}}$). Minimum impedance $Z_{\min} = R$, maximum current $I_{\max} = (V / R)$. Quality factor $Q = (\omega_0 L / R) = (1 / \omega_0 C R) = (1 / R) \sqrt{L / C} = (\omega_0 / \Delta\omega)$.

■■ **Caution / JEE Pitfall:** At resonance, V_L and V_C are NOT zero! In fact, $V_L = V_C = Q \cdot V_{\text{supply}}$, which can be substantially higher than supply voltage (voltage magnification)!

■ **Speed Trick:** Resonance bandwidth $\Delta\omega = (R / L)$. To sharpen resonance (higher Q), decrease resistance R or increase the (L / C) ratio.

Representative Source PYQs:

- **[2024 Shift] Q#3 (MCQ):** "Which of the following combinations should be selected for better tuning of an L-C-R circuit used for communication?..." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "In an L-C-R series circuit, the value of only capacitance C is varied. The resulting variation of resonance frequency f_0 as a funct..." **[Ans: C]**
- **[2023 Shift] Q#2 (MCQ):** "Total impedance of a series L - C - R circuit varies with angular frequency of the AC source connected to it as shown in the graph. ..." **[Ans: N/A]**

3.4 Reactance: Inductive & Capacitive (Moderate Yield — 5 PYQs)

Core Principle: Pure Inductor: Voltage leads current by 90° ($(\pi / 2)$), inductive reactance $X_L = \omega L = 2\pi f L$. Pure Capacitor: Current leads voltage by 90° , capacitive reactance $X_C = (1 / \omega C) = (1 / 2\pi f C)$.

■■ **Caution / JEE Pitfall:** In direct current (DC, $f = 0$), an inductor offers zero reactance ($X_L = 0$, short circuit) while a capacitor offers infinite reactance ($X_C \rightarrow \infty$, blocking DC completely).

■ **Speed Trick:** Mnemonic 'CIVIL': In a Capacitor (C), Current (I) leads Voltage (V); in an Inductor (L), Voltage (V) leads Current (I).

Representative Source PYQs:

- **[2025 Shift] Q#4 (MCQ):** "In domestic electric mains supply, the voltage and the current are..." **[Ans: N/A]**
- **[2022 Shift] Q#1 (MCQ):** "In a series LCR circuit, $R=300\Omega, L=0.9H, C=2.0\mu F$ and $\omega=1000\text{rad/s}$, then impedance of the circuit is..." **[Ans: N/A]**
- **[2020 Shift] Q#1 (MCQ):** "The power factor of R-L circuit is $\frac{1}{\sqrt{3}}$. If the inductive reactance is 2Ω . The value of resistance is..." **[Ans: N/A]**

3.5 LC Oscillations & Transformers (Moderate Yield — 4 PYQs)

Core Principle: Ideal transformer equations: $(V_s / V_p) = (N_s / N_p) = (I_p / I_s)$. Efficiency $\eta = \frac{P_{out}}{P_{in}} = (V_s I_s / V_p I_p)$. LC tank oscillations: charge $q(t) = q_0 \cos(\omega_0 t)$ with total energy conserved $U = (q^2 / 2C) + (1/2) L I^2$.

■ **Caution / JEE Pitfall:** Transformers operate exclusively on AC; in DC, primary flux is static ($(d\Phi / dt) = 0$), producing zero secondary voltage and potentially burning the primary coil!

■ **Speed Trick:** In step-up transformers ($N_s > N_p$), voltage increases but current decreases, keeping total power $V I$ conserved under ideal conditions.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A small town with a demand of 900kW of electric power at 220V is situated 20km away from an electric power genera..." [Ans: N/A]
- **[2023 Shift] Q#4 (MCQ):** "An ideal transformer has a turns ratio of 10. When the primary is connected to 220V, 50Hz as source, the power output is..." [Ans: N/A]
- **[2021 Shift] Q#2 (MCQ):** "LC-oscillations are similar and analogous to the mechanical oscillations of a block attached to a spring. The electrical equivalent of the f..." [Ans: N/A]

3.6 Power in AC Circuits & Power Factor (Standard Yield — 1 PYQs)

Core Principle: Average power dissipated $P_{avg} = V_{rms} I_{rms} \cos\phi$. Power factor $\cos\phi = (R / Z) = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$. Wattless current component $I_{wattless} = I_{rms} \sin\phi$.

■ **Caution / JEE Pitfall:** Power factor is $\cos\phi = (R / Z)$, NOT $\tan\phi$ or (R / X) ! For pure L or pure C, $\phi = 90^\circ$ implies $\cos\phi = 0$, so average power dissipation is strictly zero.

■ **Speed Trick:** Apparent power $S = V_{rms} I_{rms}$ (in VA). True power $P = S \cos\phi$ (in W). Reactive power $Q = S \sin\phi$ (in VAR).

Representative Source PYQs:

- **[2019 Shift] Q#1 (MCQ):** "An inductor of inductance L and resistor R are joined together in series and connected by a source of frequency ω . The power dis..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A small town with a demand of 900kW of electric power at 220V is situated 20km away from an electric power generating station. Each wire of the two-wire transmission line has a resistance per unit length of $5 \times 10^{-4} \Omega \text{m}^{-1}$. The town gets power from the line through a 45000V to 220V step-down transformer at a substation in the town. The line power loss in the form of heat is:

- (A) 4 kW
- (B) 8 kW
- (C) 40 kW
- (D) 80 kW

Correct Answer: None

Solution Analysis: The town receives power through a step-down transformer whose primary voltage is 45000V. The power demand of the town is $P = 900 \text{ kW} = 900000 \text{ W}$. So the current in the transmission line is $I = (P / V) = (900000 / 45000) = 20 \text{ A}$. Now, each wire is 20km long. $20 \text{ km} = 20000 \text{ m}$. Resistance per unit length of each wire is $5 \times 10^{-4} \Omega \text{m}^{-1}$. So resistance of one wire is $R = 5 \times 10^{-4} \times 20000 = 10 \Omega$. Since it is a two-wire transmission line, total resistance is $R_{total} = 10 + 10 = 20 \Omega$. Power loss in the line

Benchmark Problem 2: KCET 2026 (Q#2)

A light bulb rated 100 W is connected to an AC source of 220 V, 50 Hz. The rms current through the bulb is

- (A) 0.454 A
- (B) 0.545 A
- (C) 2.20 A
- (D) 0.22 A

Correct Answer: None

Solution Analysis: For a light bulb, the given power rating is $P=100\text{W}$ and the AC voltage given is the rms voltage, $V_{\text{(rms)}}=220\text{V}$. For a bulb, which behaves like a resistor, the power is $P=V_{\text{(rms)}}I_{\text{(rms)}}$. So, $I_{\text{(rms)}}=\frac{P}{V_{\text{(rms)}}}$. Substitute the values: $I_{\text{(rms)}}=(100 / 220)$ $I_{\text{(rms)}}=0.454\text{A}$. So, the rms current through the bulb is 0.454A. Correct option: Option A.

Benchmark Problem 3: KCET 2025 (Q#2)

A series LCR circuit containing an AC source of 100 V has an inductor and a capacitor of reactances 24Ω and 16Ω respectively. If a resistance of 6Ω is connected in series, then the potential difference across the series combination of inductor and capacitor only is

- (A) 80 V
- (B) 400 V
- (C) 8 V
- (D) 40 V

Correct Answer: None

Solution Analysis: $X_{\text{(C)}}=24\Omega$ $X_{\text{(L)}}=16\Omega$ $R=6\Omega$ $Z=\sqrt{(R^2)+(X_{\text{(C)}}-X_{\text{(L)}})^2}=\sqrt{(6^2)+8^2}=\sqrt{(100)}=10\Omega$ $i=(V / Z)=(100 / 10)=10\text{A}$ Net voltage across capacitor and inductor $=i(X_{\text{(C)}}-X_{\text{(L)}})=10(24-16)=10(8)=80\text{V}$

Benchmark Problem 4: KCET 2023 (Q#3)

In series LCR circuit at resonance, the phase difference between voltage and current is

- (A) Zero
- (B) π
- (C) $(\pi / 4)$
- (D) $(\pi / 2)$

Correct Answer: None

Solution Analysis: In L - C - R series resonance circuit, $X_{\text{(L)}}=X_{\text{(C)}}$ $\therefore \tan\phi=\frac{X_{\text{(L)}}-X_{\text{(C)}}}{R}=(0 / R)=0 \Rightarrow \phi=0^\circ$

Benchmark Problem 5: KCET 2022 (Q#1)

In a series LCR circuit, $R=300\Omega$, $L=0.9H$, $C=2.0\mu F$ and $\omega=1000\text{rad/s}$, then impedance of the circuit is

- (A) 900Ω
- (B) 500Ω
- (C) 400Ω
- (D) 1300Ω

Correct Answer: None

Solution Analysis: Given, $R=300\Omega$, $L=0.9H$, $\omega=1000\text{rad/s}$ $C=2.0\mu F=2 \times 10^{-6}F$ Impedance of given R - L - C circuit, $Z = \sqrt{(R^2) + (X_L - X_C)^2} = \sqrt{(R^2) + (\omega L - (1 / \omega C))^2}$ $\therefore X_L = \omega L$ and $X_C = (1 / \omega C)$ $= \sqrt{((300)^2) + (1000 \times 0.9 - \frac{1}{1000 \times 2 \times 10^{-6}})^2} = \sqrt{(90000 + (900 - 500)^2)} = \sqrt{(90000 + 160000)} = \sqrt{(250000)} = 500\Omega$